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JEE MAIN · QUESTION PAPER

JEE Main 2024 (Online) 29th January Evening Shift

JEE Main Question Paper with Answer Key

29 Jan 2024	90	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

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Answer key	at the end

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Chemistry 30 questions

Q1. MCQ (Single Correct Answer)

On passing a gas, 'X', through Nessler's reagent, a brown precipitate is obtained. The gas 'X' is

(A) Cl_2 (B) CO_2 (C) NH_3 (D) H_2S

Open solution

Q2. MCQ (Single Correct Answer)

Match List I with List II

	List - I (Bio Polymer)		List - II (Monomer)
(A)	Starch	(I)	nucleotide
(B)	Cellulose	(II)	α-glucose
(C)	Nucleic acid	(III)	β-glucose
(D)	Protein	(IV)	α-amino acid

Choose the correct answer from the options given below:

(A) A-IV, B-II, C-I, D-III (B) A-I, B-III, C-IV, D-II

(C) A-II, B-I, C-III, D-IV (D) A-II, B-III, C-I, D-IV

Open solution

Q3. MCQ (Single Correct Answer)

Which of the following acts as a strong reducing agent? (Atomic number: $Ce = 58, Eu = 63, Gd = 64, Lu = 71$)

(A) Eu^{2+} (B) Gd^{3+} (C) Lu^{3+} (D) Ce^{4+}

Open solution

Q4. MCQ (Single Correct Answer)

Given below are two statements:

Statement I : Fluorine has most negative electron gain enthalpy in its group.

Statement II : Oxygen has least negative electron gain enthalpy in its group.

In the light of the above statements, choose the most appropriate from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

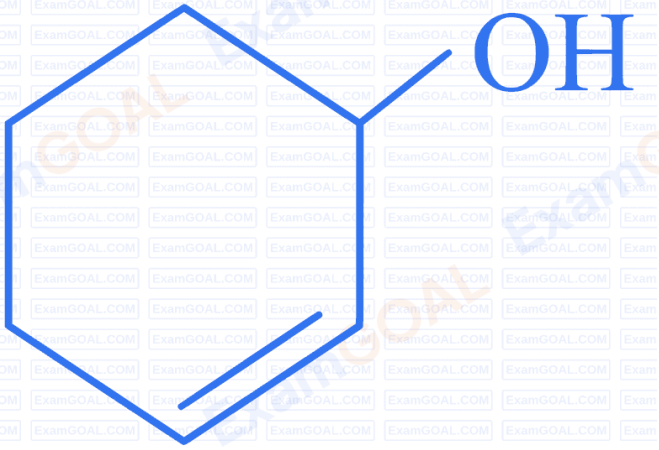
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Q5. MCQ (Single Correct Answer)

Open solution

According to IUPAC system, the compound



is named as

(A) Cyclohex-2-en-1-ol

(B) 1-Hydroxyhex-2-ene

(C) Cyclohex-1-en-3-ol

(D) Cyclohex-1-en-2-ol

Q6. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List - I (Spectral Series for Hydrogen)		List - II (Spectral Region/Higher Energy State)
(A)	Lyman	(I)	Infrared region
(B)	Balmer	(II)	UV region
(C)	Paschen	(III)	Infrared region
(D)	Pfund	(IV)	Visible region

Choose the correct answer from the options given below:

(A) A-II, B-IV, C-III, D-I

(B) A-I, B-III, C-II, D-IV

(C) A-II, B-III, C-I, D-IV

(D) A-I, B-II, C-III, D-IV

Q7. MCQ (Single Correct Answer)

Open solution

The element having the highest first ionization enthalpy is

(A) C

(B) Al

(C) Si

(D) N

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Q8. MCQ (Single Correct Answer)

Open solution

Which of the following statements are correct about Zn, Cd and Hg ?

A. They exhibit high enthalpy of atomization as the d-subshell is full.

B. Zn and Cd do not show variable oxidation state while Hg shows $+\mathrm{I}$ and $+\mathrm{II}$.

C. Compounds of Zn, Cd and Hg are paramagnetic in nature.

D. Zn, Cd and Hg are called soft metals.

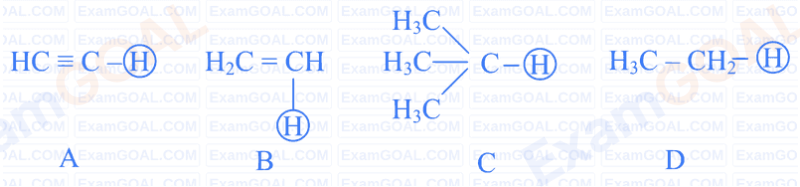
Choose the most appropriate from the options given below:

(A) C, D only **(B)** B, C only **(C)** A, D only **(D)** B, D only

Q9. MCQ (Single Correct Answer)

Open solution

The ascending acidity order of the following H atoms is



(A) A < B < D < C **(B)** A < B < C < D **(C)** D < C < B < A **(D)** C < D < B < A

Q10. MCQ (Single Correct Answer)

Open solution

The correct IUPAC name of $\mathrm{K_2MnO_4}$ is

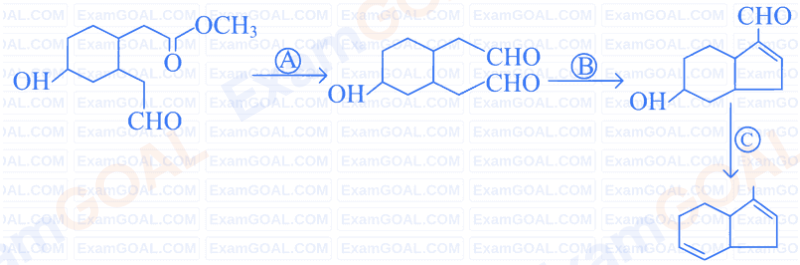
(A) Potassium tetraoxidomanganate (VI) **(B)** Dipotassium tetraoxidomanganate (VII)

(C) Potassium tetraoxopermanganate (VI) **(D)** Potassium tetraoxidomanganese (VI)

Q11. MCQ (Single Correct Answer)

Open solution

Identify the reagents used for the following conversion



(A) $\mathrm{A = LiAlH_4,\ B = NaOH_{(aq)}, C = NH_2 - NH_2/KOH}$, ethylene glycol

(B) $\mathrm{A = DIBAL - H, B = NaOH_{(aq)}, C = NH_2 - NH_2/KOH}$, ethylene glycol

(C) $\mathrm{A = LiAlH_4,\ B = NaOH_{(alc)}, C = Zn/HCl}$

(D) $\mathrm{A = DIBAL - H, B = NaOH_{(alc)}, C = Zn/HCl}$

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Q12. MCQ (Single Correct Answer)

Open solution

Chromatographic technique/s based on the principle of differential adsorption is / are

A. Column chromatography

B. Thin layer chromatography

C. Paper chromatography

Choose the most appropriate answer from the options given below:

(A) B only **(B)** A only **(C)** A & B only **(D)** C only

Q13. MCQ (Single Correct Answer)

Open solution

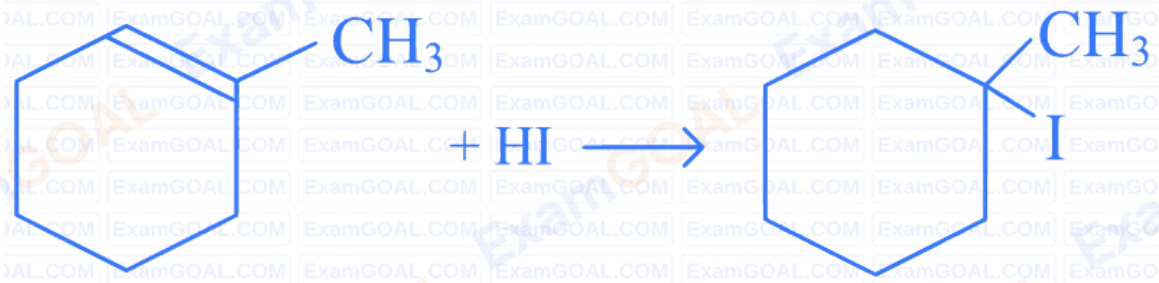
Which of the following reaction is correct?

(A) $C_2H_5CONH_2 + Br_2 + NaOH \rightarrow C_2H_5CH_2NH_2 + Na_2CO_3 + NaBr + H_2O$

(B)



(C)



(D) $CH_3CH_2CH_2NH_2 \xrightarrow[H_2O]{HNO_2, 0^\circ C} CH_3CH_2OH + N_2 + HCl$

Q14. MCQ (Single Correct Answer)

Open solution

Anomalous behavior of oxygen is due to its

(A) large size and low electronegativity **(B)** small size and high electronegativity

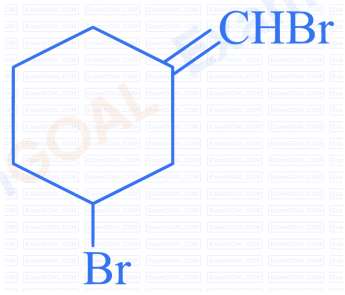
(C) small size and low electronegativity **(D)** large size and high electronegativity

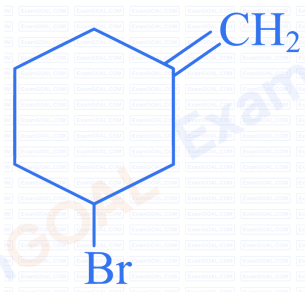
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Q15. MCQ (Single Correct Answer)

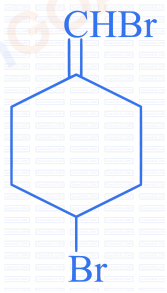
Open solution

Which one of the following will show geometrical isomerism?

(A) 

(B) 

(C) 

(D) 

Q16. MCQ (Single Correct Answer)

Open solution

A reagent which gives brilliant red precipitate with Nickel ions in basic medium is

(A) dimethyl glyoxime

(B) sodium nitroprusside

(C) meta-dinitrobenzene

(D) neutral $FeCl_3$

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Q17. MCQ (Single Correct Answer)

Phenol treated with chloroform in presence of sodium hydroxide, which further hydrolyzed in presence of an acid results

Open solution

(A) Salicylic acid

(B) Benzene-1,2-diol

(C) 2-Hydroxybenzaldehyde

(D) Benzene-1,3-diol

Q18. MCQ (Single Correct Answer)

Match List I with List II

Open solution

	List - I (Compound)		**List - II (pK_a value)**
(A)	Ethanol	(I)	10.0
(B)	Phenol	(II)	15.9
(C)	m-Nitrophenol	(III)	7.1
(D)	p-Nitrophenol	(IV)	8.3

Choose the correct answer from the options given below:

(A) A-I, B-II, C-III, D-IV

(B) A-IV, B-I, C-II, D-III

(C) A-II, B-I, C-IV, D-III

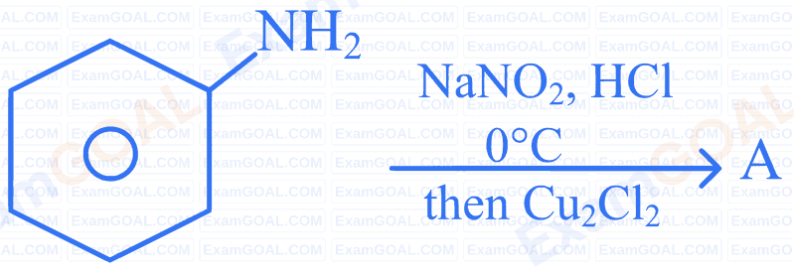
(D) A-III, B-IV, C-I, D-II

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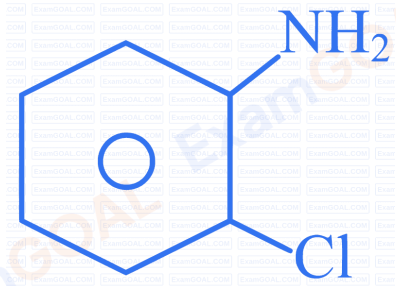
Q19. MCQ (Single Correct Answer)

Open solution

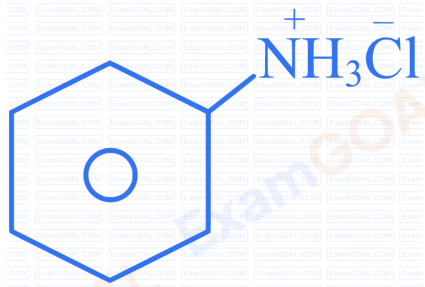
The product A formed in the following reaction is



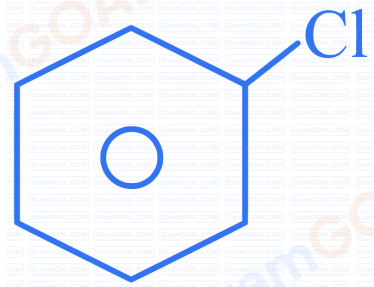
(A)



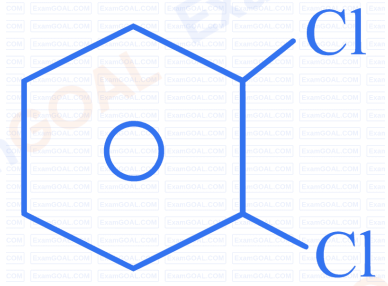
(B)



(C)



(D)



Q20. MCQ (Single Correct Answer)

Open solution

Alkyl halide is converted into alkyl isocyanide by reaction with

(A) KCN (B) NH_4CN (C) $NaCN$ (D) $AgCN$

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Q21. Numerical

The total number of molecules with zero dipole moment among $CH_4, BF_3, H_2O, HF, NH_3, CO_2$ and SO_2 is _____.

Open solution

Q22. Numerical

Standard enthalpy of vapourisation for CCl_4 is $30.5\ kJ\ mol^{-1}$. Heat required for vapourisation of $284\ g$ of CCl_4 at constant temperature is _____ kJ.

(Given molar mass in $gmol^{-1}$; $C = 12, Cl = 35.5$)

Open solution

Q23. Numerical

If $50\ mL$ of $0.5M$ oxalic acid is required to neutralise $25\ mL$ of $NaOH$ solution, the amount of $NaOH$ in $50\ mL$ of given $NaOH$ solution is _____ g.

Open solution

Q24. Numerical

The following concentrations were observed at $500\ K$ for the formation of NH_3 from N_2 and H_2. At equilibrium ; $[N_2] = 2 \times 10^{-2}M, [H_2] = 3 \times 10^{-2}M$ and $[NH_3] = 1.5 \times 10^{-2}M$. Equilibrium constant for the reaction is _____.

Open solution

Q25. Numerical

A constant current was passed through a solution of $AuCl_4^-$ ion between gold electrodes. After a period of 10.0 minutes, the increase in mass of cathode was $1.314\ g$. The total charge passed through the solution is _____ $\times 10^{-2}\ F$.

(Given atomic mass of $Au = 197$)

Open solution

Q26. Numerical

The total number of anti bonding molecular orbitals, formed from $2s$ and $2p$ atomic orbitals in a diatomic molecule is _____.

Open solution

Q27. Numerical

The half-life of radioisotope bromine - 82 is 36 hours. The fraction which remains after one day is _____ $\times 10^{-2}$.

(Given antilog $0.2006 = 1.587$)

Open solution

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Q28. Numerical

The total number of 'Sigma' and 'Pi' bonds in 2-formylhex-4-enoic acid is _____.

Open solution

Q29. Numerical

The oxidation number of iron in the compound formed during brown ring test for NO_3^- ion is _____.

Open solution

Q30. Numerical

Molality of 0.8 M H_2SO_4 solution (density 1.06 g cm^{-3}) is _____ $\times 10^{-3}$ m.

Open solution

Mathematics 30 questions

Q31. MCQ (Single Correct Answer)

Number of ways of arranging 8 identical books into 4 identical shelves where any number of shelves may remain empty is equal to

(A) 18 (B) 16 (C) 12 (D) 15

Open solution

Q32. MCQ (Single Correct Answer)

Let A be the point of intersection of the lines $3x + 2y = 14, 5x - y = 6$ and B be the point of intersection of the lines $4x + 3y = 8, 6x + y = 5$. The distance of the point $P(5, -2)$ from the line AB is

(A) $\frac{13}{2}$ (B) 8 (C) $\frac{5}{2}$ (D) 6

Open solution

Q33. MCQ (Single Correct Answer)

If $\int \frac{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x \sin(x-\theta)}} dx = A\sqrt{\cos\theta \tan x - \sin\theta} + B\sqrt{\cos\theta - \sin\theta \cot x} + C$, where C is the integration constant, then AB is equal to

(A) $2\sec\theta$ (B) $8\,\text{cosec}(2\theta)$ (C) $4\,\text{cosec}(2\theta)$ (D) $4\sec\theta$

Open solution

Q34. MCQ (Single Correct Answer)

The distance of the point $(2, 3)$ from the line $2x - 3y + 28 = 0$, measured parallel to the line $\sqrt{3}x - y + 1 = 0$, is equal to

(A) $3 + 4\sqrt{2}$ (B) $6\sqrt{3}$ (C) $4 + 6\sqrt{3}$ (D) $4\sqrt{2}$

Open solution

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Q35. MCQ (Single Correct Answer)

Let a unit vector $\hat{u} = x\hat{i} + y\hat{j} + z\hat{k}$ make angles $\frac{\pi}{2}, \frac{\pi}{3}$ and $\frac{2\pi}{3}$ with the vectors $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}, \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ respectively. If $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ then $|\hat{u} - \vec{v}|^2$ is equal to

(A) $\frac{11}{2}$ (B) $\frac{5}{2}$ (C) 7 (D) 9

Open solution

Q36. MCQ (Single Correct Answer)

The function $f(x) = \frac{x}{x^2-6x-16}, x \in \mathbb{R} - \{-2, 8\}$

(A) decreases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(B) increases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(C) decreases in $(-2, 8)$ and increases in $(-\infty, -2) \cup (8, \infty)$

(D) decreases in $(-\infty, -2)$ and increases in $(8, \infty)$

Open solution

Q37. MCQ (Single Correct Answer)

If R is the smallest equivalence relation on the set $\{1, 2, 3, 4\}$ such that $\{(1, 2), (1, 3)\} \subset \mathrm{R}$, then the number of elements in R is _____.

(A) 15 (B) 10 (C) 12 (D) 8

Open solution

Q38. MCQ (Single Correct Answer)

If the mean and variance of five observations are $\frac{24}{5}$ and $\frac{194}{25}$ respectively and the mean of the first four observations is $\frac{7}{2}$, then the variance of the first four observations is equal to

(A) $\frac{5}{4}$ (B) $\frac{4}{5}$ (C) $\frac{105}{4}$ (D) $\frac{77}{12}$

Open solution

Q39. MCQ (Single Correct Answer)

Let $\mathrm{P}(3, 2, 3), \mathrm{Q}(4, 6, 2)$ and $\mathrm{R}(7, 3, 2)$ be the vertices of $\triangle\mathrm{PQR}$. Then, the angle $\angle\mathrm{QPR}$ is

(A) $\cos^{-1}\left(\frac{7}{18}\right)$ (B) $\frac{\pi}{6}$

(C) $\cos^{-1}\left(\frac{1}{18}\right)$ (D) $\frac{\pi}{3}$

Open solution

Q40. MCQ (Single Correct Answer)

Let $y = \log_e\left(\frac{1-x^2}{1+x^2}\right), -1 < x < 1$. Then at $x = \frac{1}{2}$, the value of $225\,(y' - y'')$ is equal to

(A) 732 (B) 736 (C) 742 (D) 746

Open solution

Q41. MCQ (Single Correct Answer)

If $\log_e \mathrm{a}, \log_e \mathrm{b}, \log_e \mathrm{c}$ are in an A.P. and $\log_e \mathrm{a} - \log_e 2\,\mathrm{b}, \log_e 2\,\mathrm{b} - \log_e 3\mathrm{c}, \log_e 3\mathrm{c} - \log_e \mathrm{a}$ are also in an A.P, then $a : b : c$ is equal to

(A) $6 : 3 : 2$ (B) $9 : 6 : 4$ (C) $25 : 10 : 4$ (D) $16 : 4 : 1$

Open solution

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Q42. MCQ (Single Correct Answer)

If $\sin\left(\frac{y}{x}\right) = \log_e|x| + \frac{\alpha}{2}$ is the solution of the differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x$ and $y(1) = \frac{\pi}{3}$, then α^2 is equal to

(A) 12 (B) 9 (C) 4 (D) 3

Open solution

Q43. MCQ (Single Correct Answer)

An integer is chosen at random from the integers $1, 2, 3, \ldots, 50$. The probability that the chosen integer is a multiple of atleast one of 4, 6 and 7 is

(A) $\frac{8}{25}$ (B) $\frac{9}{50}$ (C) $\frac{14}{25}$ (D) $\frac{21}{50}$

Open solution

Q44. MCQ (Single Correct Answer)

Let $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = 12\vec{a} + 4\vec{b}$ and $\overrightarrow{OC} = \vec{b}$, where O is the origin. If S is the parallelogram with adjacent sides OA and OC, then $\frac{\text{area of the quadrilateral OA BC}}{\text{area of S}}$ is equal to _____.

(A) 7 (B) 6 (C) 8 (D) 10

Open solution

Q45. MCQ (Single Correct Answer)

The function $f(x) = 2x + 3(x)^{\frac{2}{3}}, x \in \mathbb{R}$, has

(A) exactly one point of local minima and no point of local maxima

(B) exactly one point of local maxima and exactly one point of local minima

(C) exactly two points of local maxima and exactly one point of local minima

(D) exactly one point of local maxima and no point of local minima

Open solution

Q46. MCQ (Single Correct Answer)

If each term of a geometric progression $a_1, a_2, a_3, \ldots$ with $a_1 = \frac{1}{8}$ and $a_2 \neq a_1$, is the arithmetic mean of the next two terms and $S_n = a_1 + a_2 + \ldots . + a_n$, then $S_{20} - S_{18}$ is equal to

(A) -2^{15} (B) 2^{15} (C) -2^{18} (D) 2^{18}

Open solution

Q47. MCQ (Single Correct Answer)

Let r and θ respectively be the modulus and amplitude of the complex number $z = 2 - i\left(2\tan\frac{5\pi}{8}\right)$, then (r, θ) is equal to

(A) $\left(2\sec\frac{11\pi}{8}, \frac{11\pi}{8}\right)$ (B) $\left(2\sec\frac{3\pi}{8}, \frac{3\pi}{8}\right)$

(C) $\left(2\sec\frac{5\pi}{8}, \frac{3\pi}{8}\right)$ (D) $\left(2\sec\frac{3\pi}{8}, \frac{5\pi}{8}\right)$

Open solution

Q48. MCQ (Single Correct Answer)

The sum of the solutions $x \in \mathbb{R}$ of the equation $\frac{3\cos 2x + \cos^3 2x}{\cos^6 x - \sin^6 x} = x^3 - x^2 + 6$ is

(A) 3 (B) 1 (C) 0 (D) -1

Open solution

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Q49. MCQ (Single Correct Answer)

Let $x = \frac{m}{n}$ (m, n are co-prime natural numbers) be a solution of the equation $\cos\left(2\sin^{-1}x\right) = \frac{1}{9}$ and let $\alpha, \beta(\alpha > \beta)$ be the roots of the equation $mx^2 - nx - m + n = 0$. Then the point (α, β) lies on the line

(A) $3x - 2y = -2$ (B) $3x + 2y = 2$ (C) $5x + 8y = 9$ (D) $5x - 8y = -9$

Open solution

Q50. MCQ (Single Correct Answer)

Let $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$. The sum of the prime factors of $\left|P^{-1}AP - 2I\right|$ is equal to

(A) 66 (B) 27 (C) 23 (D) 26

Open solution

Q51. Numerical

Let O be the origin, and M and N be the points on the lines $\frac{x-5}{4} = \frac{y-4}{1} = \frac{z-5}{3}$ and $\frac{x+8}{12} = \frac{y+2}{5} = \frac{z+11}{9}$ respectively such that MN is the shortest distance between the given lines. Then $\overrightarrow{OM} \cdot \overrightarrow{ON}$ is equal to _____.

Open solution

Q52. Numerical

Let the area of the region $\{(x, y) : 0 \le x \le 3, 0 \le y \le \min\{x^2 + 2, 2x + 2\}\}$ be A. Then 12 A is equal to _____.

Open solution

Q53. Numerical

Let the set $C = \{(x, y) \mid x^2 - 2^y = 2023, x, y \in \mathbb{N}\}$. Then $\sum_{(x,y)\in C} (x + y)$ is equal to _____.

Open solution

Q54. Numerical

Let the slope of the line $45x + 5y + 3 = 0$ be $27r_1 + \frac{9r_2}{2}$ for some $r_1, r_2 \in \mathbb{R}$. Then $\lim\limits_{x\to 3}\left(\int_3^x \frac{8t^2}{\frac{3r_2x}{2} - r_2x^2 - r_1x^3 - 3x}dt\right)$ is equal to _____.

Open solution

Q55. Numerical

Let α, β be the roots of the equation $x^2 - \sqrt{6}x + 3 = 0$ such that $\mathrm{Im}(\alpha) > \mathrm{Im}(\beta)$. Let a, b be integers not divisible by 3 and n be a natural number such that $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a + ib), i = \sqrt{-1}$. Then $n + a + b$ is equal to _____.

Open solution

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Q56. Numerical

Let for any three distinct consecutive terms a, b, c of an A.P, the lines $ax + by + c = 0$ be concurrent at the point P and $Q(\alpha, \beta)$ be a point such that the system of equations

$x + y + z = 6$,
$2x + 5y + \alpha z = \beta$ and
$x + 2y + 3z = 4$, has infinitely many solutions. Then $(PQ)^2$ is equal to _____.

Open solution

Q57. Numerical

Let $P(\alpha, \beta)$ be a point on the parabola $y^2 = 4x$. If P also lies on the chord of the parabola $x^2 = 8y$ whose mid point is $\left(1, \frac{5}{4}\right)$, then $(\alpha - 28)(\beta - 8)$ is equal to _____.

Open solution

Q58. Numerical

Remainder when $64^{32^{32}}$ is divided by 9 is equal to _____.

Open solution

Q59. Numerical

Let $f(x) = \sqrt{\lim\limits_{r \to x} \left\{ \frac{2r^2[(f(r))^2 - f(x)f(r)]}{r^2 - x^2} - r^3 e^{\frac{f(r)}{r}} \right\}}$ be differentiable in $(-\infty, 0) \cup (0, \infty)$ and $f(1) = 1$. Then the value of ea, such that $f(a) = 0$, is equal to _____.

Open solution

Q60. Numerical

If $\int\limits_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1 - \sin 2x}\, dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3}$, where α, β and γ are rational numbers, then $3\alpha + 4\beta - \gamma$ is equal to _____.

Open solution

Physics 30 questions

Q61. MCQ (Single Correct Answer)

A small liquid drop of radius R is divided into 27 identical liquid drops. If the surface tension is T, then the work done in the process will be:

(A) $4\pi \mathrm{R}^2\ \mathrm{T}$ (B) $8\pi R^2\ \mathrm{T}$ (C) $\frac{1}{8}\pi R^2 T$ (D) $3\pi R^2\ \mathrm{T}$

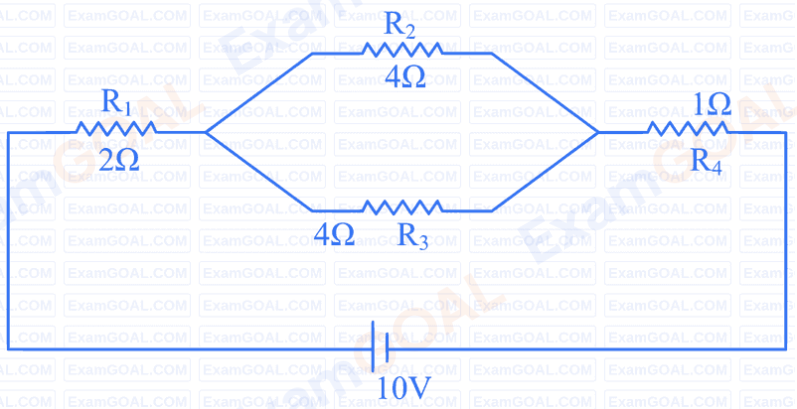
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Q62. MCQ (Single Correct Answer)

Open solution

In the given circuit, the current in resistance R_3 is :



(A) 2 A (B) 1.5 A (C) 1 A (D) 2.5 A

Q63. MCQ (Single Correct Answer)

Open solution

A physical quantity Q is found to depend on quantities a, b, c by the relation $Q = \frac{a^4b^3}{c^2}$. The percentage error in a, b and c are $3\%, 4\%$ and 5% respectively. Then, the percentage error in Q is :

(A) 43% (B) 34% (C) 66% (D) 14%

Q64. MCQ (Single Correct Answer)

Open solution

A stone of mass $900\ \text{g}$ is tied to a string and moved in a vertical circle of radius $1\ \text{m}$ making $10\ \text{rpm}$. The tension in the string, when the stone is at the lowest point is (if $\pi^2 = 9.8$ and $g = 9.8\ \text{m/s}^2$) :

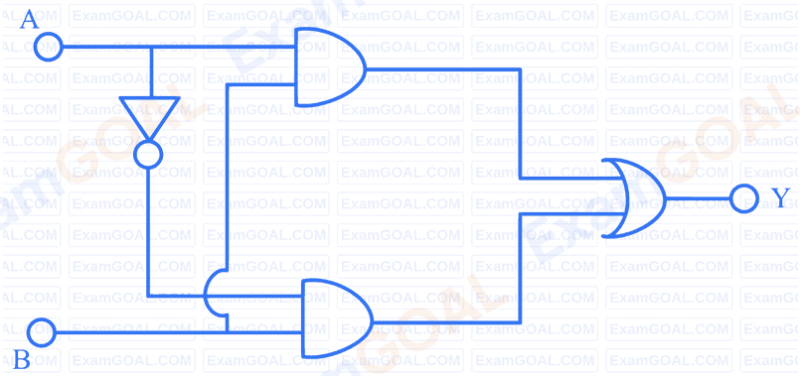
(A) 17.8 N (B) 97 N (C) 9.8 N (D) 8.82 N

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Q65. MCQ (Single Correct Answer)

Open solution

The truth table for this given circuit is :



(A)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(B)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(C)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

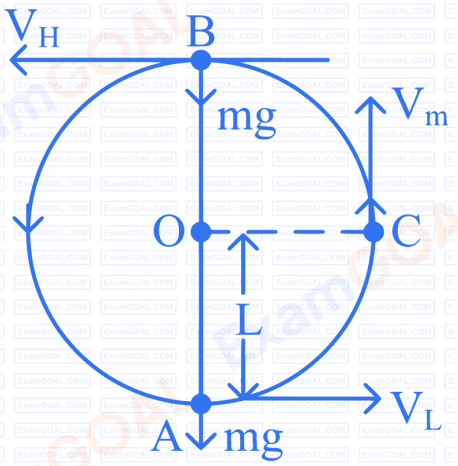
(D)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

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Q66. MCQ (Single Correct Answer)

A bob of mass 'm' is suspended by a light string of length 'L'. It is imparted a minimum horizontal velocity at the lowest point A such that it just completes half circle reaching the top most position B. The ratio of kinetic energies $\frac{(K.E)_A}{(K.E)_B}$ is :



Open solution

(A) 5 : 1 (B) 3 : 2 (C) 1 : 5 (D) 2 : 5

Q67. MCQ (Single Correct Answer)

An electric field is given by $(6\hat{i}+5\hat{j}+3\hat{k})\mathrm{N/C}$. The electric flux through a surface area $30\hat{i}\,\mathrm{m}^2$ lying in YZ-plane (in SI unit) is :

(A) 60 (B) 90 (C) 180 (D) 150

Open solution

Q68. MCQ (Single Correct Answer)

A planet takes 200 days to complete one revolution around the Sun. If the distance of the planet from Sun is reduced to one fourth of the original distance, how many days will it take to complete one revolution :

(A) 20 (B) 50 (C) 100 (D) 25

Open solution

Q69. MCQ (Single Correct Answer)

A wire of length L and radius r is clamped at one end. If its other end is pulled by a force F, its length increases by l. If the radius of the wire and the applied force both are reduced to half of their original values keeping original length constant, the increase in length will become:

(A) 2 times (B) 4 times (C) 3 times (D) $\frac{3}{2}$ times

Open solution

Q70. MCQ (Single Correct Answer)

The temperature of a gas having 2.0×10^{25} molecules per cubic meter at $1.38\,\mathrm{atm}$ (Given, $\mathrm{k}=1.38\times10^{-23}\mathrm{JK}^{-1}$) is :

(A) 500 K (B) 300 K (C) 200 K (D) 100 K

Open solution

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Q71. MCQ (Single Correct Answer)

If the distance between object and its two times magnified virtual image produced by a curved mirror is $15\ \text{cm}$, the focal length of the mirror must be:

(A) -10 cm (B) -12 cm (C) 15 cm (D) 10/3 cm

Open solution

Q72. MCQ (Single Correct Answer)

A plane electromagnetic wave of frequency $35\ \text{MHz}$ travels in free space along the X-direction. At a particular point (in space and time) $\vec{E} = 9.6\hat{j}\ \text{V/m}$. The value of magnetic field at this point is :

(A) $9.6\hat{j}T$ (B) $3.2 \times 10^{-8}\hat{i}T$ (C) $9.6 \times 10^{-8}\hat{k}T$ (D) $3.2 \times 10^{-8}\hat{k}T$

Open solution

Q73. MCQ (Single Correct Answer)

In an a.c. circuit, voltage and current are given by:

$V = 100\sin(100t)V$ and $I = 100\sin\left(100t + \frac{\pi}{3}\right)\text{mA}$ respectively.

The average power dissipated in one cycle is:

(A) 5 W (B) 25 W (C) 2.5 W (D) 10 W

Open solution

Q74. MCQ (Single Correct Answer)

A particle is moving in a straight line. The variation of position 'x' as a function of time 't' is given as $x = (t^3 - 6t^2 + 20t + 15)m$. The velocity of the body when its acceleration becomes zero is :

(A) 6 m/s (B) 10 m/s (C) 8 m/s (D) 4 m/s

Open solution

Q75. MCQ (Single Correct Answer)

The bob of a pendulum was released from a horizontal position. The length of the pendulum is $10\ \text{m}$. If it dissipates 10% of its initial energy against air resistance, the speed with which the bob arrives at the lowest point is:

[Use, g : $10\ \text{ms}^{-2}$]

(A) $5\sqrt{6}\ \text{ms}^{-1}$ (B) $5\sqrt{5}\ \text{ms}^{-1}$ (C) $2\sqrt{5}\ \text{ms}^{-1}$ (D) $6\sqrt{5}\ \text{ms}^{-1}$

Open solution

Q76. MCQ (Single Correct Answer)

N moles of a polyatomic gas $(f = 6)$ must be mixed with two moles of a monoatomic gas so that the mixture behaves as a diatomic gas. The value of N is :

(A) 6 (B) 2 (C) 4 (D) 3

Open solution

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Q77. MCQ (Single Correct Answer)

In Young's double slit experiment, light from two identical sources are superimposing on a screen. The path difference between the two lights reaching at a point on the screen is $7\lambda/4$. The ratio of intensity of fringe at this point with respect to the maximum intensity of the fringe is :

Open solution

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Q78. MCQ (Single Correct Answer)

Given below are two statements:

Statement I : Most of the mass of the atom and all its positive charge are concentrated in a tiny nucleus and the electrons revolve around it, is Rutherford's model.

Statement II : An atom is a spherical cloud of positive charges with electrons embedded in it, is a special case of Rutherford's model.

In the light of the above statements, choose the most appropriate from the options given below

Open solution

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both statement I and statement II are false

Q79. MCQ (Single Correct Answer)

Two sources of light emit with a power of $200\ \mathrm{W}$. The ratio of number of photons of visible light emitted by each source having wavelengths $300\ \mathrm{nm}$ and $500\ \mathrm{nm}$ respectively, will be :

Open solution

(A) $5:3$ (B) $3:5$ (C) $1:5$ (D) $1:3$

Q80. MCQ (Single Correct Answer)

Two particles X and Y having equal charges are being accelerated through the same potential difference. Thereafter they enter normally in a region of uniform magnetic field and describe circular paths of radii R_1 and R_2 respectively. The mass ratio of X and Y is :

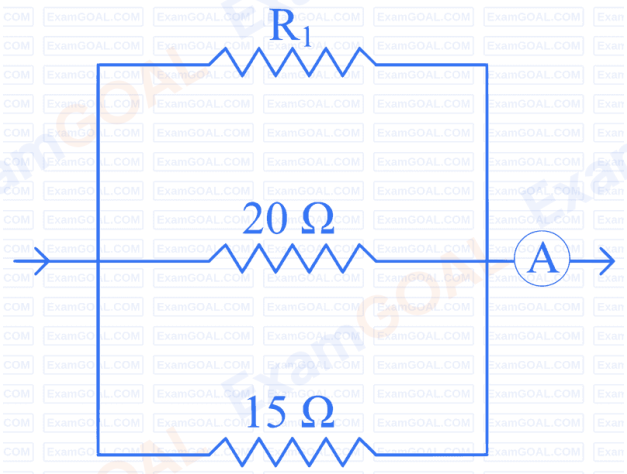
Open solution

(A) $\left(\frac{R_1}{R_2}\right)$ (B) $\left(\frac{R_2}{R_1}\right)$ (C) $\left(\frac{R_2}{R_1}\right)^2$ (D) $\left(\frac{R_1}{R_2}\right)^2$

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Q81. Numerical

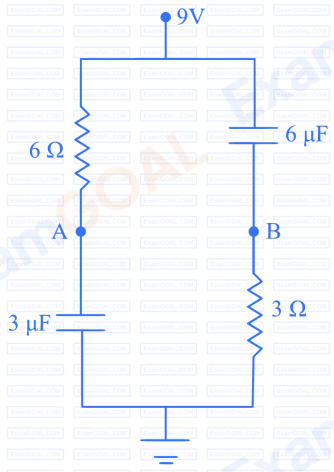
In the given circuit, the current flowing through the resistance 20Ω is 0.3 A, while the ammeter reads 0.9 A. The value of R_1 is _____ Ω.



Open solution

Q82. Numerical

In the given figure, the charge stored in $6\mu F$ capacitor, when points A and B are joined by a connecting wire is _____ μC.



Open solution

Q83. Numerical

In a single slit diffraction pattern, a light of wavelength $6000\overset{o}{A}$ is used. The distance between the first and third minima in the diffraction pattern is found to be 3 mm when the screen in placed 50 cm away from slits. The width of the slit is _____ $\times 10^{-4}$ m.

Open solution

Q84. Numerical

A body of mass 5 kg moving with a uniform speed $3\sqrt{2}\,\text{ms}^{-1}$ in $X-Y$ plane along the line $y = x + 4$. The angular momentum of the particle about the origin will be _____ $\text{kg m}^2\,\text{s}^{-1}$.

Open solution

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Q85. Numerical

Hydrogen atom is bombarded with electrons accelerated through a potential difference of V, which causes excitation of hydrogen atoms. If the experiment is being performed at $\mathrm{T} = 0\,\mathrm{K}$, the minimum potential difference needed to observe any Balmer series lines in the emission spectra will be $\frac{\alpha}{10}\,\mathrm{V}$, where $\alpha =$ _____.

Open solution

Q86. Numerical

A simple harmonic oscillator has an amplitude A and time period 6π second. Assuming the oscillation starts from its mean position, the time required by it to travel from $x =$ A to $x = \frac{\sqrt{3}}{2}$ A will be $\frac{\pi}{x}\,\mathrm{s}$, where $x =$ _____.

Open solution

Q87. Numerical

Two metallic wires P and Q have same volume and are made up of same material. If their area of cross sections are in the ratio $4:1$ and force F_1 is applied to P, an extension of Δl is produced. The force which is required to produce same extension in Q is F_2.

The value of $\frac{F_1}{F_2}$ is _____.

Open solution

Q88. Numerical

A particle is moving in a circle of radius $50\,\mathrm{cm}$ in such a way that at any instant the normal and tangential components of it's acceleration are equal. If its speed at $\mathrm{t} = 0$ is $4\,\mathrm{m/s}$, the time taken to complete the first revolution will be $\frac{1}{\alpha}\left[1 - e^{-2\pi}\right]\mathrm{s}$, where $\alpha =$ _____.

Open solution

Q89. Numerical

A charge of $4.0\mu\mathrm{C}$ is moving with a velocity of $4.0 \times 10^6\,\mathrm{ms}^{-1}$ along the positive y axis under a magnetic field $\vec{B}$ of strength $(2\hat{k})\mathrm{T}$. The force acting on the charge is $x\hat{i}N$. The value of x is _____.

Open solution

Q90. Numerical

A horizontal straight wire $5\,\mathrm{m}$ long extending from east to west is falling freely at right angle to horizontal component of earths magnetic field $0.60 \times 10^{-4}\,\mathrm{Wbm}^{-2}$. The instantaneous value of emf induced in the wire when its velocity is $10\,\mathrm{ms}^{-1}$ is _____ $\times 10^{-3}\,\mathrm{V}$.

Open solution

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Answer Key

Chemistry

1. **C**	2. **D**	3. **A**	4. **C**	5. **A**
6. **A**	7. **D**	8. **D**	9. **D**	10. **A**
11. **D**	12. **C**	13. **C**	14. **B**	15. **A**
16. **A**	17. **C**	18. **C**	19. **C**	20. **D**
21. **3**	22. **56**	23. **4**	24. **417**	25. **2**
26. **4**	27. **63**	28. **22**	29. **1**	30. **815**

Mathematics

31. **D**	32. **D**	33. **B**	34. **C**	35. **B**
36. **A**	37. **B**	38. **A**	39. **D**	40. **B**
41. **B**	42. **D**	43. **D**	44. **C**	45. **B**
46. **A**	47. **B**	48. **D**	49. **C**	50. **D**
51. **9**	52. **164**	53. **46**	54. **12**	55. **49**
56. **113**	57. **192**	58. **1**	59. **2**	60. **6**

Physics

61. **B**	62. **C**	63. **B**	64. **C**	65. **B**
66. **A**	67. **C**	68. **D**	69. **A**	70. **A**
71. **A**	72. **D**	73. **C**	74. **C**	75. **D**
76. **C**	77. **A**	78. **B**	79. **B**	80. **D**
81. **30**	82. **36**	83. **2**	84. **60**	85. **121**
86. **2**	87. **16**	88. **8**	89. **32**	90. **3**

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